

NEW DEVELOPMENT FOR HYUNDAI REFS. FORD: GE2B1

Offer number: OFFER 16022023SAT_HYUNDAI refs. 21026-21027-21127_v10
Date: 16/02/2023



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Carretera a Pesquería-Los Ramones
KMS 13 al 15 Interior 18
"La Arena" Pesquería C.P.66679,
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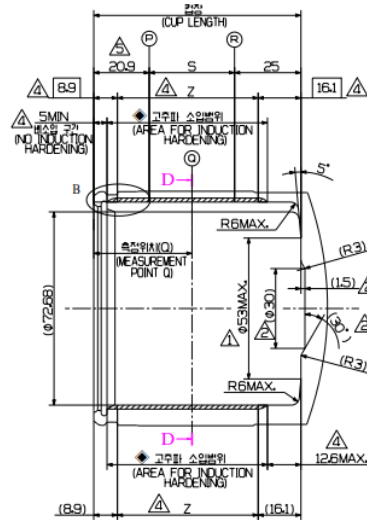
TECHNICAL OFFER

NEW DEVELOPMENT FOR HYUNDAI REFS.

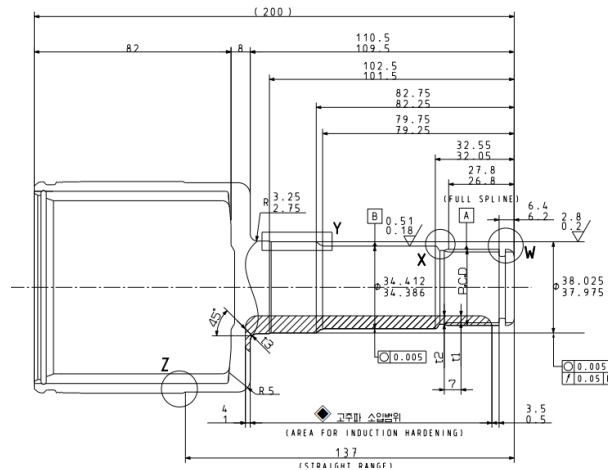
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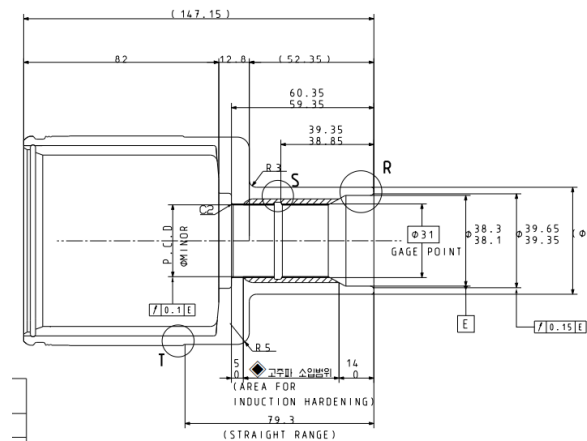
- 21027-ST700 drawing n° 21027_ST700_HDJ27_ORACE_CUP_R6_w



- 21027-GE701 drawing n° 21027-GE701_HDJ27_MALE_V22C0067_220725_w



- 21127-GE701 drawing n° 21127-GE701_HDJ27_FEMALE_V22C0067_220725_w



No modifications in the tempering tunnel of the 3207 GH machine are needed for the treatment.

1.2. Upgrading and development description

As mentioned in the introduction to this document, we proceed to describe the modifications and tasks necessary for the adaptation of the OF:3206 and OF:3207 machines to carry out the treatment of the mentioned references.

MECHANICAL MODIFICATIONS

According to the customer's needs, the GH mechanical Department describes the needs and also the tasks necessary to comply with the hardening specifications indicated by HYUNDAI.

All necessary modifications are detailed below:

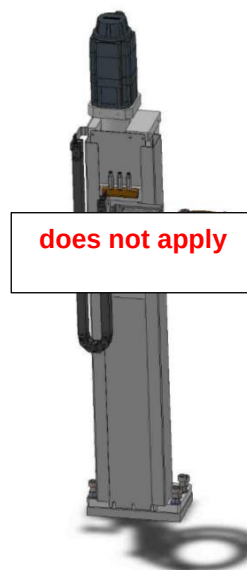
Manipulation modifications:

- CURVED DISC ON INLET CONVEYOR
- ENTRANCE CONVEYOR STOPPER - WAY TO WASHING MACHINE
- SUPPLEMENT PLATES GRIPPER FINGERS - WASHING MACHINE INPUT MANIPULATOR
- GRIPPER FINGER SUPPLEMENT PLATES - WASHER OUTLET MANIPULATOR
- 4 PLATE TOOLS - TYPE 482-605-00190.
These are the four rotary plate jigs needed to hold the new references. In the current tools it is not possible to place the new parts.
- PLATE OF CLAWS. GEARED MOTOR 12.04.731. VARIOUS PARTS OF ASSEMBLY 482.205.00005
It is necessary to install a geared motor in the chuck since currently it is not necessary to rotate the tulips under treatment, but for the new references it will be necessary.
- GRIPPER FINGER SUPPLEMENT PLATES - EDDY CURRENT MANIPULATOR AND MARKER
Due to the different diameter of the upper parts of the tulips, the claws must be modified by adding some supplements to be able to handle the parts.
- SEVERAL LITTLE MATERIAL
When we talk about small material, we refer to all the small components, pneumatics, fittings, screws and small sensors to be able to detect the rotation of the geared motor to be installed.

Column Servo-driven for quench (OF:3206): **does not apply**

~~In accordance with the need for the bowl of female part, 21027-ST700, GH considers in this offer the installation of this additional column to carry out the cooling in the 3206 machine.~~

- ~~— COLUMN SERVO DRIVEN~~
- ~~— SEVERAL LITTLE MATERIAL~~



Eddy current coils:

This offer includes the price for the new bowls coils. For stems is not necessary new coils. The current ones are valid.

This offer addresses the need for new eddy current coils for the bowls only. Therefore, as each coil comprises (1 for mater and 1 for inspection), 2 sets of coils will be necessary in the installation.

- COILS SET FOR BOWLS.
21026_ST000E_BJ_ORACE_CUP_V22C0055 2 units. (1 mater + 1 inspection)
21027_ST700_HDJ27_ORACE_CUP_R6 2 units. (1 mater + 1 inspection)

SOFTWARE MODIFICATIONS

According to the customer's needs, the GH Software and programming Department describes the needs and also the tasks necessary to comply with the hardening specifications indicated by HYUNDAI.

Laser marking modifications

What would be done is to be able to send the X,Y coordinate where the marking is done from the PLC program. If there were more changes to the marking to be made (data to be marked, format, etc...) or something had to be changed in the mechanical sequence for the marking, it would NOT be included.

Hyundai request/clarification:

Hyundai WIA just wants to modify/improve in interfacing so that they can call the marking program (which was/will be created/made in laser marker for different marking position) on the main OP of GH induction machine.

On the main OP of GH induction machine, they want to make it possible select/interwork the corresponding marking program according to the reference part.

For example, if the #1 reference part is selected the corresponding #1 marking program should be automatically called/interworked.

They know the height of marking lens should be manually adjusted by operator, and Hyundai WIA is talking about the interfacing between PLC (GH machine) and PC (Keyence Laser marker)
(Currently, Hyundai WIA has to modify the marking program every time only in PC of laser marker (when changing over to another reference parts which is required to set different marking position) as they cannot call another marking program on main OP. And they have some issue in the quality of marking. Sometimes the size of marking get different and sometimes the marking is not done in proper position by improper modification done by operator)

CNC Servo addition for quench column **does not apply**

~~This point considers the hardware part necessary for the addition of the column quench. Necessary for the treatment of the bowl of female part. (21027-ST700)~~

~~For the new column and the vertical treatment to be carried out on the new references, it is necessary to add the servomotor to carry out the movement of the external quench while heating the inductor. As the piece is the one that is moving, the quench must be controlled by means of a servomotor.~~

~~It would be necessary to add an additional axis with an additional axis license (not auxiliary) to be able to move it with a G01 with the Z axis. We should consider the additional license if necessary, apart from checking space, capacity of the power supply and new module. The shower is still the existing one, so it is not contemplated to add any additional curve in the IPM~~

- ~~• Servo 1FK7042-2AF71-1RB2 + cables are added.~~
- ~~• The power supply is enough to add a 1FK7042 servo, so we keep it.~~

- ~~Double power module, since we do not have space in the cabinet to add a single power module. The idea is to replace one of the simple modules that they currently have, with a double module.~~
- ~~Additional axis license.~~

Turning on the hardening claw of the tulip (PLATE OF CLAWS. GEARED MOTOR 12.04.731)

This point includes the control elements necessary for the installation of the geared motor mentioned in the mechanical section for the rotation of the parts. Now the treatment is done statically, but when moving to movement these two hardware elements are necessary.

This point considers the hardware part necessary for the controlled turning of the claw. This will be able to clamp the stem of the parts.

Hardware material. It will imply adding a variator and contemplating the rotation or not depending on the selected recipe

- Frequency inverter.
- motor guard.

HEAT TREATMENT AND DEVELOPMENT

According to the customer's needs, the GH treatment applications department describes the needs and also the tasks necessary to comply with the hardening specifications indicated by HYUNDAI.

For the part 21026-ST000E:

- This part will be treated in the OF: 3207 GH machine **by single shot**.
- For the treatment of this bowl, GH presents in this offer the inductor necessary for its treatment. It also features the necessary external quench.
- The inclusion in this offer of the necessary manufacturing tools will be common for all inductors, which will only be included in the cost the first time the inductor is manufactured.
- **IMPORTANT:** No laboratory tests will be performed at GH for the Bowl.

For the part 21027-ST700:

- This part will be treated in the OF: 3206 GH machine **by scanning**.
- For the treatment of this track, GH presents in this offer the inductor necessary for its treatment. No additional external quench will be necessary.
- The inclusion in this offer of the necessary manufacturing tools will be common for all inductors and centering gauge, which will only be included in the cost the first time the inductor is manufactured.
- GH will include in this offer the cost of developing the application and testing in the GH laboratory.
- **Internal quench below new inductor.**
- **IMPORTANT:** The client requests to make this piece by scanning, GH cannot guarantee good results in the deformation of the piece.

For the part 21026-GE003:

- This part will be treated in the OF: 3207 GH machine **by single shot**.
- For the treatment of this stem, GH presents in this offer the inductor necessary for its treatment. No additional external quench will be necessary.
- The inclusion in this offer of the necessary manufacturing tools will be common for all inductors and centering gauge, which will only be included in the cost the first time the inductor is manufactured.
- GH will include in this offer the cost of developing the application and testing in the GH laboratory.

For the part 21027-GE701:

- This part will be treated in the OF: 3206 GH machine **by single shot**.
- For the treatment of this stem, GH presents in this offer the inductor necessary for its treatment. No additional external quench will be necessary.
- The inclusion in this offer of the necessary manufacturing tools will be common for all inductors and centering gauge, which will only be included in the cost the first time the inductor is manufactured.
- GH will include in this offer the cost of developing the application and testing in the GH laboratory.

For the part 21127-GE701:

- This part will be treated in the OF: 3206 GH machine **by single shot**.
- For the treatment of this stem, GH presents in this offer the inductor necessary for its treatment. An external quench attached to the inductor will be necessary
- The inclusion in this offer of the necessary manufacturing tools will be common for all inductors and centering gauge, which will only be included in the cost the first time the inductor is manufactured.
- GH will include in this offer the cost of developing the application and testing in the GH laboratory.

1.3. Required Plant Supplies (GH Requirements).

GH considers it necessary for the 3206 and 3207 machines to be in a perfect state of maintenance in order to proceed with the necessary tasks and modifications.

Changes in power, frequency, connection with respect to HYUNDAI facilities will not be necessary.

1.4. Technical Assistance and commissioning

In accordance with HYUNDAI's previous requirements, GH details below the scope of the implementation of these references in machines 3206-3207.

HYUNDAI requirements:

Please see the enclosed drawings and provide the quotation for all necessary components including traveling cost of GH technician(s) and transport cost of test-piece.

~~*Hyundai WIA asked us to consider "two additional separate visits" for technical support of Event and SOP as below*~~

~~*Technical support for Event: June 2023 (2 weeks)*~~

- ~~*Technical support for SOP: October 2024 (2 weeks)*~~

Based on Hyundai's comments, GH is listed in this offering as follows:

- Technical assistance for Starting-up process for 5 references (2 weeks)

For this technical assistance GH consider the intervention of the necessary personnel in Hyundai facilities. GH consider.

2 weeks GHMX Service Engineer
2 weeks GHE Engineer Programmer
2 weeks GHE Mechanical specialist

- Technical support for SOP (2 weeks) (date to be confirmed)

1.5. Transport and part delivery

According to the requirements of the reform, GH will request **30 parts** of each of the five references to HYUNDAI, which must be in charge of packaging and sending the pieces from its facilities to GH Spain.

These parts are necessary for their development and laboratory tests at the GH facilities.

This offer quotes the cost of sending these parts to GH Spain

2. Services and others

2.1. Standards

The equipment's design and manufacturing will be performed according to the "GH Standards" which can be downloaded at the following link <http://www.ghinduction.com/wp-content/uploads/2018/10/DOC-12-GH-STANDARDS-MANUFACTURE-OF-MACHINERY-ENG.pdf>

In case of having specific customer standards, mention here , followed by this text:

The customer fully agrees these "GH Standards" which are in any case complementary to the client's standards or specifications.

2.2. Final Acceptance

The final acceptance tests will be carried out after commissioning at the customer's premises.

The scope of these tests will be identical to the Pre-Acceptance tests which were carried out before delivery at GH's premises.

COMMERCIAL OFFER

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3. Pricing Proposal

According to the information provided by HWM (HYUNDAI WIA MEXICO), GH Induction propose the following offer which consists in new inductor development/reference

Scope of Supply and Price List (No Taxes Included)

APPLICATION

Code	Item	Qty.	Price (EUR)
1.	Reference 21026-ST000E		
	Hardening inductor	2	14.140 €
	Quench	1	1.700 €
	Manufacturing tool	1	600 €
2.	Reference 21027-ST700		
	Hardening inductor	2	22.480 € 23.480 €
	Manufacturing tool	1	790 €
	Manufacturing gauge	1	500 €
	Development	1	7.215 €
3.	Reference 21026-GE003		
	Hardening inductor	2	14.540 €
	Manufacturing tool	1	590 €
	Manufacturing gauge	1	320 €
	Development	1	7.215 €
4.	Reference 21027-GE701		
	Hardening inductor	2	15.400 €
	Manufacturing tool	1	590 €
	Manufacturing gauge	1	320 €
	Development	1	7.215 €
5.	Reference 21127-GE701		
	Hardening inductor	2	15.400 €
	Manufacturing tool	1	590 €
	Manufacturing gauge	1	315 €
	Development	1	7.215 €
	Laboratory tests	1	3.690 €
Total items 1 to 5, EXW Valencia (Spain)			120.825,00-€ 121.825,00-€

MECHANICAL AND SOFTWARE

Code	Item	Qty.	Price (EUR)
6.	Mechanical modifications in 3206 GH machine	1	45.905,00-€ 20.067,00-€
7.	Eddy current coils for bowls (no necessary other eddy current coils)	1	16.945,00-€
8.	Software and Hardware modifications in 3206-3207 GH machine	1	34.320,00-€ 31.460,00-€
Total items 6 to 8, EXW Valencia (Spain)			97.170,00-€ 68.472,00-€

PARTS TRANSPORT

Code	Item	Qty.	Price (EUR)
9.	Parts transport from Korea to Spain	1	900,00-€
10.	FOB Valencia Port	1	700,00-€
Total items 9 to 10, EXW Valencia (Spain)			1.600,00-€

ASSISTANCE AND STARTING-UP

Code	Item	Qty.	Price (EUR)
11.	Technical starting-up for 5 references (2 weeks). 1 additional week of GHMX Service Engineer for testing verification process.	1	53.900,00-€ 47.250,00-€
12.	Technical support for SOP: October 2024 (2 weeks)	1	17.800,00-€ 17.105,00-€
Total items 11 to 12, EXW Valencia (Spain)			89.500,00-€ 76.405,00-€

TOTAL Price NEW DEVELOPMENT FOR HYUNDAI REFS. FORD: GE2B1	256.250,00-€
SPECIAL DISCOUNT: 4%	-10.250,00-€
FINAL PRICE NEW DEVELOPMENT FOR HYUNDAI REFS. FORD: GE2B1	246.000,00-€

4. Supply conditions

Delivery Terms

Delivery Time

The system proposed by GH Induction is scheduled to be delivered as follow:

- **12 working weeks** after receipt of formal purchase order, down payment and clarification of all technical details.

Delivery Conditions

Our equipment is delivered according to the following 2010 Incoterms:

- FOB GH ELECTROTERMIA, S.A.U., San Antonio de Benagéber, València (SPAIN).

Packing Conditions

Wrapped and protected for ground transportation.

Payment Terms

100%	<i>At the delivery of the materials by bank transfer (T/T), 90 days net</i>
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Delayed Payment

GH Induction reserves its right to apply a financial interest charge on any delayed payment beyond agreed and stipulated date. This charge of interest has a value of 6% (value 2015) and is calculated on the entire due amount. Calculation of such financial interest shall be made by sections of 14 days minimum.

Non-payment or delayed payment of our invoices - for any reason whatsoever - gives us the indisputable right to demand advance payment for the remaining supply, without prejudice of any other action, or to consider the contract suspended or annulled, and to annul the practice of any other contracts that may be in force, without giving the buyer the right to any type of refund or compensation for damages, or other.

Pricing revision

During the validity of our offer or after an order our pricing can be reviewed if one of the following terms occurs:

- Customer Parts designs change.
- Change in standards definition after end of design period.
- Change in machine's scope.

Offer Validity

This is a firm offer with prices valid for a period of 60 days

Warranty

- GH guarantees its products, while and to the exceptions listed below, against any manufacturing defect or defects in the materials used. This warranty GH requires repair or replacement, in their view, of any part recognized as defective, without compensation by any of the parties.
The items damaged during the warranty period should be referred to GH freight prepaid for examination and repair or replace GH prepaid by ordinary transport. Destination custom tariffs will never be paid by GH.
- Excluded from this warranty:
 - Damaged parts for wear and tear, negligent handling, inadequate oversight and excessive overloads.
 - Modifications, repairs or assembly and commissioning of new developments not made or authorized by GH.
 - The parts imposed by the buyer.
 - The elements to be explicitly specified such as inductors, tools, etc. (any wear elements).

- The labour assembly and disassembly of defective parts.
- The total duration of the warranty is 12 months from the acceptance of the machine or up to 15 months from delivery (unless a different period is specified in the purchase agreement).
- The guarantee does not imply that all replacement parts are available at the nearest centre GH. For this reason, we invite the client to purchase the recommended spares.
- GH will not be liable for damages due to delay in providing services in the work resulting from causes beyond the reasonable control of GH. These causes include, without limitation, compliance with any regulations, governmental orders or instructions, natural disasters, acts or omissions of Customer, acts of civil or military authorities, fires, strikes, war, riot or delay in transportation. GH in no event be liable for special or consequential damages caused by delay due to any reason.
- The guarantee is applied at the place where the sale is made or at the end point indicated in the contract.

Equipment Property

The equipment remains the property of GH Induction until payment is duly performed. However, once the equipment is transferred to customer's plant responsibility is transferred to user and the user's insurance will cover the entire value of the said machine.

Any controversy, claim or dispute arising between the parties about the interpretation and fulfilment of this sales contract will be referred to the exclusive jurisdiction of the Courts of Valencia, Spain. This sales contract is likewise exclusively governed by Spanish Law.

Design Property and Process Know-How

Machine(s) design(s) remain the property of GH Induction Group. No detail drawings will be provided unless specific agreement which does not allow, anyway the use of the machine design for machine manufacturing. Such behaviour would be considered as an infringement of copyright.

The following design won't be provided:

- Detailed mechanical drawings
- Inductor drawings
- Electronic control boards
- Dies or tooling designs

Assembly drawings will be provided for proper management of the maintenance operation

In addition to those designs, the customer understands the importance of the know-how and commercial information we are sharing while providing commercial or technical offers. The client decisively undertakes not to disclose any technical or commercial information or any other data which he might learn about during the contractual relationship. In the event of any breach of the above on the part of the client, the seller reserves the right to claim compensation for damages.

Order Cancellation

In case where the order would have to be cancelled before machine's production termination, GH Induction would be entitled to a fair compensation. This compensation would be following some general guidelines:

- At design level: 15% of the entire project value.
- After design and prior final machine assembly:
 - Value of finished goods increased by 1.40 applied on book value for handling cost and financial compensation.
- During machine assembly and before machine shipment:
 - 100% of contract value.

Once this payment is made, the components or finished goods will be fully transferred to the customer.

Use of Customer's References

GH Induction will be, unless clearly mentioned in the contract, entitled to use customer's name as reference within its reference list for its promotion. Of course, no data or information about handled process will be provided to any third party.



Personal Data Protection

Your details will become part of a database owned by GH ELECTROTERMIA, S.A.U. and processed for the purposes of CUSTOMER MANAGEMENT, MAINTENANCE, PAYMENTS, DISPATCH OF COMMERCIAL COMMUNICATIONS AND NEWSLETTERS.

You also expressly agree that GH ELECTROTERMIA, S.A.U. may use your personal details to send, by any means, including electronic means, commercial information and advertising on its products and services.

Customer details will be processed at all times pursuant to Regulation (EU) 2016/679, on Personal Data Protection.

You may exercise your rights to access, rectification, opposition, deletion and limitation of treatment by writing to the database manager at the following address: C/. Vereda Real s/n, 46184 San Antonio de Benagéber (València - SPAIN).